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RANGE SENSING AND SAFETY SYSTEM FOR A LUMINAIRE

Abstract

An automated luminaire and method are provided. The automated luminaire includes a lens position sensor in a lens positioning mechanism of the automated luminaire and a control system. The lens position sensor is separate from sensors in the lens positioning mechanism and senses an actual lens position of the lens positioning mechanism. The control system reduces an emitted beam power density when the emitted light beam has an actual beam angle different than a beam angle commanded by the control system. The method includes determining, based on a value of a lens position sensor separate from other sensors in an associated lens positioning mechanism, whether the emitted light beam has a beam angle commanded by a control system of the automated luminaire and reducing an emitted beam power density of the emitted light beam when the beam angle is different than the beam angle commanded by the control system.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 18/801,149 filed Aug. 12, 2024 by Michal Stanek, et al., entitled “RANGE SENSING AND SAFETY SYSTEM FOR A LUMINAIRE,” which claims the benefit of U.S. Provisional Application No. 63/579,736 filed Aug. 30, 2023, entitled “RANGE SENSING AND SAFETY SYSTEM FOR A LUMINAIRE,” both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD OF THE DISCLOSURE

[0002] The disclosure generally relates to luminaires, and more specifically to a range sensing and safety system for an automated luminaire.

BACKGROUND

[0003] Some luminaires in the entertainment and architectural lighting markets include automated and remotely controllable functions. Such luminaires may be used in theatres, television studios, concerts, theme parks, night clubs and other venues. A luminaire may provide control over the pan and tilt functions of the luminaire allowing the operator to control the direction the luminaire is pointing and thus the position of the light beam on the stage or in the studio. Such position control may be obtained via control of the luminaire's position in two orthogonal rotational axes, which may be referred to as pan and tilt. Some luminaires provide control over other parameters such as intensity, color, focus, beam size, beam shape, and/or beam pattern.

[0004] The optical systems of such automated luminaires may be designed to enable a user to control the beam size, from a very narrow output beam to a wider, wash beam. Such control may allow such luminaires to be used with long throws to a target or for almost parallel light effects, as well as for wider, more traditional wash effects. Optical systems with the ability to produce narrow beams may be referred to as ‘Beam’ optics, while optical systems with the ability to produce wide beams may be referred to as ‘Wash’ optics.

SUMMARY

[0005] In a first embodiment, an automated luminaire includes a lens position sensor in a lens positioning mechanism of the automated luminaire and a control system. The lens position sensor is configured to sense an actual lens position of the lens positioning mechanism, wherein the lens position sensor is separate from sensors in the lens positioning mechanism. The control system is configured to determine, based on a value of the lens position sensor, whether an emitted light beam from the automated luminaire has an actual beam angle different than a commanded beam angle commanded by the control system and reduce an emitted beam power density of the emitted light beam when the emitted light beam has an actual beam angle different than a commanded beam angle commanded by the control system.

[0006] In a second embodiment, a method of controlling an intensity of a beam emitted by an automated luminaire includes determining, based on a value of a lens position sensor separate from

other sensors in an associated lens positioning mechanism, whether the emitted light beam has a beam angle commanded by a control system of the automated luminaire and reducing an emitted beam power density of the emitted light beam when the beam angle is different than the beam angle commanded by the control system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in conjunction with the accompanying drawings in which like reference numerals indicate like features.

[0008] FIG. 1 presents an isometric view of a luminaire according to the disclosure;

[0009] FIG. 2A presents an isometric view of a head of the luminaire of FIG. 1 with a range sensing module according to the disclosure attached to the head;

[0010] FIG. 2B presents an exploded isometric view of the head of FIG. 2A with the range sensing module detached from the head;

[0011] FIG. 3 presents an exploded isometric view of the range sensing module according to the disclosure;

[0012] FIG. 4 presents a diagrammatic representation of a range sensing module according to the disclosure in use; and

[0013] FIG. 5 presents a flow chart of a process for controlling intensity of a light beam emitted from a luminaire according to the disclosure.

DETAILED DESCRIPTION

[0014] Preferred embodiments are illustrated in the figures, like numerals being used to refer to like and corresponding parts of the various drawings.

[0015] Some luminaires (both automated and non-automated) comprise a light source including a discharge lamp or a light emitting diode (LED) array with converging optics. Luminaires according to the disclosure comprise one or more laser based light sources, where a laser LED is used as a pump for a light emitting phosphor. The light source for such a system is designed to be small, with a reduced etendue. In some embodiments, a light source according to the disclosure includes a laser-based light engine that comprises one or more lasers in one or more colors and one or more optical devices that convert a coherent light beam emitted by the laser into an incoherent light beam. Such lasers may include a solid state LED laser. The optical devices may comprise phosphor panels or filters. In some such embodiments, the light engine comprises a solid state LED laser emitting coherent blue light that excites a phosphor to emit incoherent white light.

[0016] Luminaires comprising a light source according to the disclosure may comprise a variety of optical systems: a fixed “beam” optical system that produces a narrow beam, a fixed “wash” optical system that produces a wide beam, or a zoom optical system that can change the beam diameter from a narrow beam to a wide beam. Such a zoom optical system may also have the ability to produce good quality imagery from gobos or patterns inserted into the optical train.

[0017] Luminaires emitting a very narrow beam (e.g., from a laser-based light source) may emit a light beam having a high emitted beam power density. If the emitted beam power density exceeds certain predetermined threshold power density value, such a beam may cause damage to the eyes or body of persons in the beam, or damage to scenery, structure, or fabrics. For this reason, luminaires emitting such light beams may be subject to various regulatory controls in various jurisdictions.

[0018] While the light output from such laser-based light sources is not coherent laser light, it may still have a high emitted beam power density, as discussed above. Luminaires according to the disclosure include methods and systems for satisfying regulatory requirements and mitigating the potential physical problems of an emitted beam of high power density.

[0019] FIG. 1 presents an isometric view of a luminaire **100** according to the disclosure. The luminaire **100** is an automated luminaire comprising a head **128** which is coupled to a tilt mechanism to rotate within a yoke **120** about a tilt axis **124**. The yoke **120** is coupled to a pan mechanism to rotate relative to a fixed enclosure **126** about a pan axis **122**. The pan axis **122** and the tilt axis **124** are orthogonal to each other. An operator of the luminaire **100** is thus able to direct light output from the automated luminaire in a desired direction, through rotation of the head **128** about the pan axis **122** and tilt axis **124**. The yoke **120** and the head **128** may be mechanically coupled for rotation about the pan axis **122** and tilt axis **124**, respectively, to hand-operated manual controls or may be coupled for motion to motors, linear actuators, or other electromechanically controlled mechanisms for control of an orientation of the yoke **120** and the head **128**.

[0020] Such electromechanical mechanisms may be electrically coupled to a control system (or controller) **102** of the luminaire **100**, the control system **102** configured to control a motion of the electromechanical mechanisms and thus the orientation of the yoke **120** and the head **128**. In various embodiments, the control system **102** comprises a microcontroller or other programmable processing system. In some embodiments, the control system **102** may be coupled for local control to a user interface **104** included in the luminaire **100**, and configured to receive therefrom signals relating to desired orientations of the yoke **120** and the head **128**.

[0021] In other embodiments, the control system **102** may be coupled for remote control by a data link (wired or wireless) to a remotely located control console, to receive signals therefrom indicating a desired orientation of the head **128**. The data link may use DMX512 (Digital Multiplex) protocol or other suitable communication protocol, e.g., Art-Net, ACN (Architecture for Control Networks), and Streaming ACN. In such embodiments, the control system **102** is configured to move the pan and/or tilt mechanisms in response to signals received via the data link. In some such embodiments, the control system **102** moves the pan mechanism in response to a control signal received on a first control channel of the data link and moves the tilt mechanism in response to a control signal received on a second control channel of the data link.

[0022] The luminaire **100** further includes a range sensing module **110** that comprises a first range sensing system **114**, electrically coupled to the control system **102**. In some embodiments, the range sensing module **110** may further include a second range sensing system **112**, also electrically coupled to the control system **102**.

[0023] FIG. 2A presents an isometric view of the head **128**, with a range sensing module **110** attached. FIG. 2B presents an exploded isometric view of the head **128**, with the range sensing module **110** detached. While the range sensing module **110** is shown as mounted to the head **128** of the automated luminaire **100**, range sensing modules (and associated luminaire control system safety procedures) according to the disclosure may be used in other embodiments with other types of luminaires (for example, automated and non-automated luminaires, as well as moving and static luminaires).

[0024] FIG. 3 presents an exploded isometric view of the range sensing module **110**. The first range sensing system **114** comprises a first range sensor **302** mounted in an aperture **303** of the range sensing module **110**. The first range sensor **302** is mounted behind a window **304**, which protects it from airborne materials and moisture. The first range sensor **302** is mounted to the range sensing module **110** by a bracket **306** and covered by a cover **308**. The second range sensing system **112** comprises similar components that are not visible in FIG. 3. The range sensing systems **112** and **114** are configured to electrically couple to the control system **102**.

[0025] In some embodiments, the range sensing systems **112** and **114** may comprise Light Detection and Ranging (LiDAR) range sensors. LiDAR is a method for determining a distance to an object using a measured time-of-flight for a light pulse traveling outwards from the sensor, reflecting off the object, and returning to the sensor. A LiDAR range sensing system is analogous with a Radar system, but uses light rather than radio waves. LiDAR range sensors such as those used in the range sensing systems **112** and **114** may have a measurement range exceeding 150 m

(meters) with an accuracy of measurement of a few millimeters. In some embodiments, the range sensing module **110** may include only one of the range sensing systems **112** and **114**.

[0026] FIG. **4** presents a diagrammatic representation of the range sensing module **110** in use. The first range sensing system **114** emits light pulses **402** continuously and receives reflected light pulses from the closest object in the direction that the luminaire **100** is currently pointed. In FIG. **4** the closest object is the object **404**. The first range sensing system **114** calculates a measured distance **406** to the object **404** from a timing between the emission of the pulse and the receipt of the reflected light pulse. The control system **102** is configured to identify the object **404** as the nearest object that a light beam emitted by the luminaire **100** might illuminate. In FIG. **4** the object **404** is shown as a person, however in various applications the object **404** may be a wall, a scenic element, a seating area, or other part of the structure or contents of a performance space.

[0027] The first range sensing system **114** continually and repeatedly reports the measured distance **406** to the control system **102**. As the luminaire moves—or as objects move in front of the luminaire **100**—the object **404** may move or be moved out of the light beam and a new object move or be moved into the light beam, resulting in the first range sensing system **114** reporting to the control system **102** a new measured distance **406** from the luminaire **100** to the new, closest object.

[0028] In some embodiments, the control system **102** is configured to scan the pan and tilt mechanisms of the luminaire **100** through all (or some) possible pan/tilt positions to determine “no-go” pan/tilt positions in which the luminaire **100** is pointed at an object having a measured distance **406** that is less than a threshold distance value. Such positions (or ranges of such positions) may be referred to as intensity-controlled ranges. In other embodiments, as discussed in more detail below, an operator of the luminaire **100** identifies intensity-controlled ranges. In various embodiments, such intensity-controlled ranges are stored in a memory of the control system **102** in a look up table or other data structure. A luminaire is considered to be in an intensity-controlled range when both its pan position is in the pan position or range of pan positions of the intensity-controlled range and its tilt position is in the tilt position or range of tilt positions of the intensity-controlled range.

[0029] The control system **102** is configured to execute safety procedures to control a light beam emitted by the luminaire **100** based on one or more of (i) a current position of the pan and tilt systems, (ii) a current emitted beam power density of the luminaire **100**, and (iii) the measured distance **406** received from the first range sensing system **114**. The current emitted beam power density of the light beam may be determined based on a current output power of the light source, losses in the optical system, and a beam angle of the emitted light beam (e.g., calculated based on a configuration of the zoom lens). Devices of the optical system (optical devices) through which the beam passes before being emitted from the luminaire **100** and whose configuration may affect the current emitted beam power density include color filters, gobos, neutral density filters, and an intensity iris.

[0030] In some embodiments, if the luminaire **100** is currently emitting light at a beam angle below a predetermined threshold angle (e.g., where a resulting power density in the beam is high enough to be subject to regulatory limits), the control system **102** is configured to reduce or disable the light output from the luminaire **100** if the measured distance **406** is less than the threshold distance value. Only if the measured distance **406** is greater than the threshold distance value will the safety procedures of the control system **102** allow full light output from the luminaire **100**.

[0031] In some embodiments, such range sensing and safety procedures operate at all times that the luminaire **100** is powered up. In this way, even during set-up, when scenery, seating area, or other parts of the structure are being prepared for performance and/or when workers are on stage, the control system **102** is configured, upon detection of an object with a measured distance **406** less than the threshold distance value, to reduce or disable the light output. A value of the threshold distance may be preset (or predetermined) based on one or both of local regulations and emitted beam power determinants (such as light source power, zoom configuration, and/or optical system

configuration). In some embodiments the threshold distance value is repeatedly calculated based on a current emitted beam power density of a beam emitted by the luminaire **100**.

[0032] Some regulations set a maximum allowable value for emitted beam power density, expressed in Watts per square meter (W/m²)—that is, a ratio of emitted beam power in Watts to an illuminated area. The value of the ratio goes down when an electrical power to the light source is reduced, when the beam is dimmed by other means, or when the beam size is increased (causing the area illuminated by the beam to increase). The beam size may increase due to either or both of an increase in beam angle or an increase in measured distance **406**. As an example, if the threshold distance value is set to 10 m when the beam angle is 1° (degree), then it may be increased to 20 m when the beam angle is 4°, and further increased to 30 m when the beam angle is 8°.

[0033] In some embodiments, the control system **102** is configured to use a look up table that provides predetermined threshold distance values keyed by current values of one or more of the emitted beam power determinants. Such an embodiment may include a look up table that groups threshold distance values by ranges of values for one or more of the emitted beam power determinants. In other embodiments, the control system **102** is configured to precalculate the predetermined threshold distance value whenever any of the emitted beam power determinants changes.

[0034] In some embodiments the range sensing module **110** may comprise a second range sensing system **112**, to provide redundancy. In some such embodiments, the control system **102** is configured to compare the measured distances **406** reported by the range sensing systems **112** and **114** and uses the closer measured distance **406**. In other such embodiments, the control system **102** is configured to use the measured distance **406** from the second range sensing system **112** if it determines that the first range sensing system **114** has failed. In still other such embodiments, if both range sensing systems **112** and **114** fail (or provide measured distances **406** that differ by more than a threshold difference), the control system **102** completely shuts down operation of the luminaire **100**, per its safety procedures.

[0035] In some embodiments, zoom and/or focus lens mechanisms of the luminaire **100** comprise one or more lens position sensors configured to sense actual lens positions of the lenses within the luminaire **100**, separate from sensors in the zoom and focus lens positioning mechanisms. In such embodiments, the control system **102** is configured to determine, from values of the one or more lens position sensors, whether the luminaire **100** is actually emitting a light beam having a beam angle commanded by the control system **102**. Such a situation might arise, for example, if a lens positioning mechanism associated with the lens position sensor has lost calibration or is not operating properly. If the actual beam angle of the emitted beam is different than the commanded beam angle, the control system **102** is configured to determine whether to reduce the emitted beam power density of the light beam emitted from the luminaire **100**.

[0036] In some embodiments, the pan and tilt positioning mechanisms (luminaire head orientation mechanisms) of the luminaire **100** similarly comprise orientation sensors configured to sense an actual orientation of the luminaire head **128**. Such orientation sensors are separate from sensors in associated luminaire head motion mechanisms. In still other embodiments, the luminaire head **128** may comprise accelerometers configured to sense an orientation of the luminaire head **128** relative to the ground and/or the performance space. In such embodiments, the control system **102** is configured to determine, from a value of the sensed actual current orientation of the luminaire head **128**, whether the luminaire head **128** is actually in an orientation commanded by the control system **102**. If the control system **102** determines that the sensed actual current orientation of the luminaire head **128** is different than the orientation commanded by the control system **102**, the control system **102** is configured to reduce the emitted beam power density of the light beam emitted from the luminaire **100** (e.g., if the pan and tilt positioning mechanisms were to fail or the luminaire head **128** were to be physically moved out of position).

[0037] The control system **102** may be configured to use the measured distance **406** from the range

sensing module **110** and any or all of the pan/tilt orientation commanded by the control system **102**, the absolute position data, and/or the accelerometer data to build a three-dimensional (3D) model of the performance space, from the point of view of the luminaire **100**. Such a 3D model may be built by the control system **102** moving the luminaire head **128** through all or a subset of its orientations and storing the measured distance **406** at some or all of those orientations. In this way, an expected distance lookup table or other data structure may be created to enable the control system **102** to determine from a commanded pan/tilt orientation an expected object distance at that orientation. Such a value of expected distance may be used to check a currently measured distance **406** from the range sensing module **110** for correctness or to check whether the 3D model is still an accurate model of the environment of the luminaire **100**.

[0038] In some embodiments, the control system **102** is configured to display such a 3D model to an operator of the luminaire **100**, for the operator's use in identifying intensity-controlled areas in the performance space—for example a nearby section of the audience—where a narrow angle light beam can never be projected, for safety reasons. In some such embodiments, the control system **102** is configured to send the 3D model to an external device, where the operator may interact with a display of the 3D model to identify intensity-controlled areas in the performance space. In such embodiments, the control system **102** is further configured to receive from the external device information identifying the “intensity-controlled” areas in the performance space and use such received information to define intensity-controlled pan/tilt positions or ranges to, as discussed above. In some such embodiments, safety procedures implemented by the control system **102** may reduce beam intensity—regardless of a currently measured distance **406** from the range sensing module **110**—when an “intensity-controlled” pan/tilt position is commanded by an operator of the luminaire **100**.

[0039] Safety procedures implemented by the control system **102** of the luminaire **100** provide a range sensing and safety system according to the disclosure using the range sensing module **110** and, in some embodiments, actual current position feedback on zoom and focus lens mechanisms, actual position feedback on pan/tilt mechanism positioning, and/or accelerometers monitoring the actual orientation of the luminaire head **128**.

[0040] FIG. 5 presents a flow chart of a process for controlling intensity of a light beam emitted from a luminaire according to the disclosure. The process begins by checking emitted beam power determinants for changed values: zoom configuration, light source power, and optical system configuration. Such a change could cause the control system **102** to reduce the emitted beam power density or to permit its return to full intensity. In steps **502**, **504**, and **506**, respectively, the control system **102** checks for a change in the configuration of the zoom lens system, the power applied to the light source, and the configuration of optical devices of the optical system. If a change in any of these emitted beam power determinants is detected, the method **500** continues in step **508**, where the control system **102** calculates and stores a minimum allowable threshold distance value for a current emitted beam power density of the beam emitted by the luminaire **100**. The threshold distance value is a minimum distance from the luminaire at which the emitted beam power density of the emitted light beam is below a predetermined threshold power density value. The control system **102** performs the calculation using current values of all of the configuration of the zoom lens system, the power applied to the light source, and the configuration of the optical devices of the optical system. As discussed below, if a distance to an object in the beam is less than the calculated threshold distance value, the control system **102** reduces the emitted beam power density of the beam to a safer, predetermined emitted beam power density.

[0041] In some embodiments, the control system **102** controls a variable power applied to the light source by current control or pulse width modulation (PWM) of voltage. In such embodiments, in step **504**, the control system **102** checks whether a value of the variable power applied to the light source has changed. In other embodiments, the control system **102** controls only whether power is applied to the light source—on/off control. In such embodiments, in step **504**, the control system

102 checks whether the light source has been powered on or powered off since a previous iteration of the method **500**.

[0042] As discussed above, regulations may mandate a maximum allowable value for emitted beam power density, expressed in W/m.^{sup.2}. The emitted beam power density goes down when an electrical power to the light source is reduced, when the beam is dimmed by a configuration of the optical system, or when the configuration of the zoom lens system increases the emitted beam size, causing the area illuminated by the beam to increase. The beam size also increases with increasing distance from the luminaire **100**. Thus, a minimum allowable threshold distance may be calculated beyond which the emitted beam is safe and below which the emitted beam is not considered safe.

[0043] Returning to FIG. 5, in step **510**, the control system **102** determines whether the measured distance **406** is greater than the stored threshold distance value. Step **510** is performed either after calculating and storing a threshold distance value in step **508** or after determining in steps **502**, **504**, and **506** that no emitted beam power determinant has changed. If the control system **102** determines in step **510** that the measured distance **406** is not greater than the stored threshold distance value, in step **512** the control system **102** reduces the emitted beam power density to obtain safer operation of the luminaire **100**. In some embodiments, the emitted beam is blocked physically by a shutter or other means. In other embodiments, the power applied to the light source is reduced and/or the emitted beam is increased in size (wider beam angle). The actions performed in step **512** result in no emitted beam or an emitted beam that is safe at any distance from the luminaire **100**. From step **512**, the control system **102** proceeds to step **502** to begin another iteration of the method **500**.

[0044] If the control system **102** determines in step **510** that the measured distance **406** is greater than the stored threshold distance value, in step **514**, the control system determines whether both pan and tilt positions are within respective “intensity-controlled” ranges, as discussed above. If both pan and tilt positions are within their respective “intensity-controlled” ranges, the method **500** proceeds to step **512**, where the control system **102** reduces the emitted beam power density to obtain safer operation of the luminaire **100**. If one or both of the pan and tilt positions are outside their respective “intensity-controlled” ranges, in step **516** the control system **102** removes any reduction in the emitted beam power density previously imposed in step **512**, permitting the luminaire to return to full emitted beam power density. Thus, two conditions are determined before the control system **102** permits the luminaire to return to its full emitted beam power density: (i) the measured distance **406** is greater than the stored threshold distance value and (ii) the pan and tilt positions are not both within their respective “intensity-controlled” ranges.

[0045] In step **516**, any or all of the following actions may be performed to permit the luminaire to return to its full emitted beam power density: a physical blockage of the emitted beam is removed, a power applied to the light source is returned to a higher, commanded level, and the optical system is moved from a safe configuration to a commanded beam angle. From step **516**, the control system **102** proceeds to step **502** to begin another iteration of the method **500**.

[0046] While only some embodiments of the disclosure have been described herein, those skilled in the art, having benefit of this disclosure, will appreciate that other embodiments may be devised which do not depart from the scope of the disclosure herein. While the disclosure has been described in detail, it should be understood that various changes, substitutions, and alterations can be made hereto without departing from the spirit and scope of the disclosure.

Claims

1. An automated luminaire comprising: a lens position sensor in a lens positioning mechanism of the automated luminaire, the lens position sensor configured to sense an actual lens position of the lens positioning mechanism, wherein the lens position sensor is separate from sensors in the lens positioning mechanism; and a control system configured to: determine, based on a value of the lens position sensor, whether an emitted light beam from the automated luminaire has an actual beam

- angle different than a commanded beam angle commanded by the control system; and reduce an emitted beam power density of the emitted light beam when the emitted light beam has an actual beam angle different than a commanded beam angle commanded by the control system.
2. The automated luminaire of claim 1, wherein the control system is configured to reduce the emitted beam power density of the emitted light beam when the actual beam angle is smaller than the commanded beam angle.
 3. The automated luminaire of claim 1, wherein the emitted light beam is an incoherent light beam emitted by a light emitting phosphor optically pumped by a laser LED.
 4. The automated luminaire of claim 1, wherein the control system is configured to reduce the emitted beam power density of the emitted light beam by altering one or more of: (i) a power applied to a light source configured to generate the emitted light beam, and (ii) a configuration of an optical system through which the emitted light beam passes before being emitted from the automated luminaire.
 5. The automated luminaire of claim 1, wherein the control system is configured to reduce the emitted beam power density of the emitted light beam by physically blocking the emitted light beam.
 6. The automated luminaire of claim 1, further comprising: an orientation sensor in a luminaire head orientation mechanism of the automated luminaire, the orientation sensor configured to sense an actual current orientation of a head of the automated luminaire, wherein the orientation sensor is separate from sensors in an associated luminaire head motion mechanism, wherein the control system is configured to: determine, based on a value of the orientation sensor, whether the head of the automated luminaire has an orientation commanded by the control system; and reduce the emitted beam power density of the emitted light beam when the orientation of the head of the automated luminaire is different than the orientation commanded by the control system.
 7. A method of controlling an intensity of an emitted light beam emitted by an automated luminaire, the method comprising: determining, based on a value of a lens position sensor separate from other sensors in an associated lens positioning mechanism, whether the emitted light beam has a beam angle commanded by a control system of the automated luminaire; and reducing an emitted beam power density of the emitted light beam when the beam angle is different than the beam angle commanded by the control system.
 8. The method of claim 7, wherein the emitted beam power density of the emitted light beam is reduced when the beam angle is smaller than the beam angle commanded by the control system.
 9. The method of claim 7, wherein reducing the emitted beam power density of the emitted light beam comprises altering one or more of: (i) a power applied to a light source configured to generate the emitted light beam, and (ii) a configuration of an optical system through which the emitted light beam passes before being emitted from the automated luminaire.
 10. The method of claim 7, wherein reducing the emitted beam power density of the emitted light beam comprises physically blocking the emitted light beam.
 11. The method of claim 7, further comprising: determining, based on a value of an orientation sensor separate from other sensors in a luminaire head orientation mechanism, whether the head of the automated luminaire has an orientation commanded by the control system; and reducing the emitted beam power density of the emitted light beam when the orientation of the head of the automated luminaire is different than the orientation commanded by the control system.
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